

Topic B5: Genes, inheritance and selection

B5.1 Inheritance

Summary

Inheritance relies on the genetic information contained in the genome being passed from one generation to the next, whether sexually or asexually. The characteristics of a living organism are influenced by the genome and its interaction with the environment.

Underlying knowledge and understanding

Learners should be familiar with the idea of heredity as the process by which genetic information is passed from one generation to the next. They should have a simple model of chromosomes, genes and DNA.

Common misconceptions

Learners commonly struggle to appreciate the physical relationships between the nucleus, genetic material, the genome, chromosomes and genes. Accurate definitions of these terms will help learners' explanations in this topic. Learners

often have well-developed (although not necessarily scientifically accurate) explanations for inheritance before undertaking GCSE study. Some examples include that intra-specific variation is as a result of defects in development or that acquired characteristics can be inherited. Care must also be taken with the concept of dominant and recessive alleles. Whether an allele is dominant or recessive does not affect the mechanism of inheritance of the allele, but is an observed pattern in the phenotype of organisms. Many learners assume that the dominant allele 'dominates' the recessive allele preventing its expression (which is not the case) or that the recessive allele is actually just an absence of the dominant allele (also not generally the case).

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM5.1i	understand and use direct proportions and simple ratios in genetic crosses	M1c
BM5.1ii	understand and use the concept of probability in predicting the outcome of genetic crosses	M2e
BM5.1iii	extract and interpret information from charts, graphs and tables	M2c and M4a

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
B5.1a	explain the following terms: gamete, chromosome, gene, allele/variant, dominant, recessive, homozygous, heterozygous, genotype and phenotype			Use of alleles to work out the phenotype of progeny.
B5.1b	describe the genome as the entire genetic material of an organism			
B5.1c	describe that the genome, and its interaction with the environment, influence the development of the phenotype of an organism	use of examples of discontinuous (e.g. eye colour) and continuous variation (e.g. weight and height)		
B5.1d	Recall that all variants arise from mutations, and that most have no effect on the phenotype, some influence phenotype and a very few determine phenotype			
B5.1e ☑	describe how genetic variants may influence phenotype: - in coding DNA by altering the activity of a protein - in non-coding DNA by altering how genes are expressed	- in coding: DNA related to mutations affecting protein structure, including active sites of enzymes - in non-coding: DNA related to stopping transcription of mRNA (use of terms promoter, transcription factor not required)		
B5.1f ☑	explain some of the advantages and disadvantages of asexual and sexual reproduction in a range of organisms	the number of live offspring per birth, how quickly the organisms can reproduce verses the need for the introduction of variation in a population caused by environmental pressures		
B5.1g	explain the terms haploid and diploid			

B5.1h	explain the role of meiotic cell division in halving the chromosome number to form gametes	that this maintains diploid cells when gametes combine and is a source of genetic variation			
B5.1i	explain single gene inheritance	the context of homozygous and heterozygous crosses involving dominant and recessive genes	M2c, M4a		Prediction of the probability of phenotype for genetic crosses. Investigation into probability by suitable example (e.g. coin toss or die roll).
B5.1j	predict the results of single gene crosses	the use of Punnett squares	M1c, M2c, M2e, M4a		
B5.1k	describe sex determination in humans using a genetic cross	the use of Punnett squares	M1c, M2c, M2e, M4a		
B5.1l	recall that most phenotypic features are the result of multiple genes rather than single gene inheritance				
B5.1m ☒	describe the development of our understanding of genetics	the work of Mendel		WS1.1a, WS1.1d, WS1.1f, WS1.1i	